Compressive Neuropathies of the Upper Extremity

Resident Small Group Handout — Dr. Barrera

I. General Principles

- Compression → demyelination (early, reversible) → axonal degeneration (late, less reversible)
- Double crush: proximal compression lowers threshold for symptoms distally
- Night symptoms common in entrapment (positioning, venous congestion)
- Provocative maneuvers reproduce symptoms by transiently increasing pressure at the compression site

II. Median Nerve

A. Carpal Tunnel Syndrome

Sites of Compression

- Transverse carpal ligament (flexor retinaculum) — primary
- Boundaries: scaphoid tubercle + trapezium (radial); hook of hamate + pisiform (ulnar)
- 9 flexor tendons + median nerve in the canal
- Palmar cutaneous branch travels superficial to TCL — spared in CTS
- Recurrent motor branch: extraligamentous (46%), subligamentous (31%), transligamentous (23%)

History

- Numbness/tingling in thumb, index, long, radial ring finger
- Night symptoms — waking, shaking out hands
- Dropping objects, difficulty with buttons/jars
- Risk factors: pregnancy, diabetes, hypothyroidism, RA, dialysis, repetitive wrist use

Physical Exam

- Thenar atrophy (late — APB, opponens pollicis)
- Two-point discrimination (>6 mm abnormal)
- Durkan's compression test (direct pressure 30 sec) — most sensitive (~87%)
- Phalen's test (wrist flexion 60 sec) — Sens ~68%
- Tinel's at carpal tunnel — Sens ~50%

CTS-6 Score (Graham et al., JHS 2006; Fowler et al., JHS 2023)

6-item weighted questionnaire derived from the Boston CTS Questionnaire. Each item is yes/no with assigned point values. Max score **26**.

#	Item	Points
1	Numbness in median nerve territory	3.5
2	Nocturnal numbness	4.0
3	Thenar weakness / atrophy	5.0
4	Positive Phalen’s test	5.0
5	Loss of two-point discrimination (≤5 mm)	4.5
6	Positive Tinel’s sign	4.0

Score	Probability of CTS	Recommendation
>12	~80%	EDX may not be needed if clinical picture is consistent
≤12	~25%	EDX recommended to confirm or rule out

- Weakly positive correlation with EDX severity (Bland grades): mean CTS-6 ~15 (mild), ~19 (moderate), ~22 (severe) (Fowler et al., JHS 2023)
- Sensitivity 84%, specificity 91% (validated when administered by medical assistants)
- **As adjunct:** strengthens clinical confidence; helps triage who truly needs EDX
- **As substitute for EDX:** reasonable for high-scoring patients (≥12) with straightforward presentation, no red flags
- **Still get EDX when:** atypical presentation, suspected concurrent pathology (cubital tunnel, polyneuropathy, cervical radiculopathy), medicolegal need, or when severity grading will change management

Electrodiagnostics

- Prolonged distal sensory latency (>3.5 ms) — earliest finding
- Prolonged distal motor latency (>4.2 ms)
- Decreased SNAP/CMAP amplitude — axonal loss (later, worse prognosis)
- Comparison studies: median vs. ulnar sensory latency across wrist

Imaging

- Ultrasound: median nerve cross-sectional area >10 mm^2 at carpal tunnel inlet
- MRI: reserved for space-occupying lesions or recurrent/persistent cases
- X-ray: hook of hamate fracture, arthritis

Non-Surgical Management

- Night splinting in neutral wrist position (most evidence-based conservative Tx)
- Activity modification — avoid sustained wrist flexion/extension
- Corticosteroid injection (1 mL dexa or methylpred, ulnar to palmaris longus): good short-term relief; diagnostic value
- Oral steroids: short-course can provide temporary relief
- Nerve gliding exercises — modest evidence, low risk
- Treat underlying cause if present (hypothyroidism, diabetes optimization, RA)

Surgical Management

- Indications: failed conservative Tx (3–6 months), constant numbness, thenar atrophy, severe EDX
- Open CTR: incision ulnar to thenar crease, divide TCL under direct vision
- Endoscopic CTR: equivalent outcomes, possibly faster return to work
- Protect: recurrent motor branch, palmar cutaneous branch, superficial palmar arch

Recurrent / Persistent CTS

- Persistent (<3 mo post-op) vs. recurrent (symptom-free interval, then return)
- Workup: repeat EDX (new baseline at 3–6 months post-op), ultrasound or MRI
- Common causes:
  - Incomplete release (most common cause of persistent symptoms)
  - Perineural fibrosis/scarring
  - Tenosynovitis
  - Wrong initial diagnosis (pronator syndrome, cervical radiculopathy, polyneuropathy)
- Revision: extended incision, complete release, consider hypothenar fat pad flap, neurolysis, synovectomy

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B. Pronator Syndrome

Sites of Compression (proximal → distal)

1. Ligament of Struthers (supracondylar process → medial epicondyle) — rare
2. Lacertus fibrosus (bicipital aponeurosis)
3. Between two heads of pronator teres — most common
4. Proximal arch of FDS

Key Differentiator from CTS

- Numbness includes **palmar cutaneous branch territory** (thenar palm) — spared in CTS
- Forearm aching with activity rather than nocturnal paresthesias
- Negative Phalen's and Tinel's at wrist

Physical Exam — Localize by Provocative Test

- Resisted pronation, elbow extended → pronator teres
- Resisted elbow flexion, forearm supinated → lacertus fibrosus
- Resisted long finger PIP flexion → FDS arch

Electrodiagnostics: Often normal or equivocal; helpful mainly to rule out CTS

Non-Surgical Management

- Activity modification (limit repetitive pronation/gripping)
- NSAIDs, short course
- Splinting: elbow at 90°, forearm neutral
- Most patients improve with 3–6 months of conservative management

Surgical: Release all potential compression sites through anterior approach

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C. Anterior Interosseous Nerve (AIN) Syndrome

- Pure motor branch: FPL, FDP to index, pronator quadratus
- **Failed "OK" sign** — pulp-to-pulp pinch becomes tip-to-tip
- No sensory loss
- Many cases are Parsonage-Turner (neuritis) → observe 3–6 months before considering surgical exploration

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III. Ulnar Nerve

A. Cubital Tunnel Syndrome

Sites of Compression

1. Arcade of Struthers (~8 cm proximal to medial epicondyle)
2. Medial intermuscular septum
3. Retrocondylar groove
4. Osborne's ligament (cubital tunnel roof) — most common
5. Between two heads of FCU
6. Anconeus epitrochlearis (accessory muscle, ~11% of population)

History

- Numbness/tingling in small finger and ulnar half of ring finger
- Worse with prolonged elbow flexion (phone, sleeping)
- Grip/pinch weakness, clumsiness
- Ask: prior elbow fracture, arthritis, nerve subluxation

Physical Exam

- Elbow flexion compression test (flex + direct pressure × 60 sec) — most sensitive
- Tinel's at cubital tunnel
- Froment sign: FPL compensates for weak adductor pollicis during key pinch
- Wartenberg sign: small finger rests in abduction (weak 3rd palmar interosseous)
- Intrinsic/hypothenar atrophy (late)
- Clawing of ring and small fingers — **ulnar paradox**: clawing is worse in distal lesions
- Check for subluxation with elbow flexion/extension

Electrodiagnostics

- Motor conduction velocity across elbow <50 m/s
- Focal conduction block at elbow
- Decreased SNAP amplitude (dorsal ulnar cutaneous — branches proximal to Guyon's canal)
- Needle EMG: fibrillations/PSWs in FDI, ADM if denervation

Imaging

- X-rays: arthritis, cubitus valgus, loose bodies
- Ultrasound: nerve cross-sectional area, dynamic subluxation
- MRI: mass lesion, signal change in nerve

Non-Surgical Management

- Elbow pad worn with pad over antecubital fossa to limit flexion (or night extension splint)
- Activity modification — avoid leaning on elbow, prolonged flexion
- Nerve gliding exercises
- NSAIDs for pain
- Most effective for mild to moderate severity without atrophy or axonal loss on EDX

Surgical Management

- **In situ decompression:** release Osborne's ligament + FCU fascia; preferred when no subluxation
- **Anterior transposition:** subcutaneous, intramuscular, or submuscular; indicated for subluxating nerve, cubitus valgus, failed in situ, revision
- **Medial epicondylectomy:** less commonly performed

B. Guyon's Canal (Ulnar Tunnel) Syndrome

3 Zones

- Zone 1 (proximal): mixed motor + sensory
- Zone 2 (deep motor branch): motor only — intrinsic weakness, no sensory loss
- Zone 3 (superficial sensory branch): sensory only

Key Differentiator from Cubital Tunnel

- Dorsal ulnar cutaneous nerve branches proximal to Guyon's canal
- If dorsal hand sensation is intact → lesion is at Guyon's, not cubital tunnel

Common Causes: Ganglion cyst (most common), hook of hamate fracture, ulnar artery thrombosis (hypothenar hammer syndrome), cycling

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IV. Radial Nerve

A. Radial Tunnel Syndrome

Sites of Compression

1. Fibrous bands anterior to radial head
2. Recurrent radial vessels (leash of Henry)
3. Tendinous margin of ECRB
4. Arcade of Frohse (proximal supinator edge) — most common
5. Distal edge of supinator

Key Point: This is a **pain syndrome, not a weakness syndrome** — distinguishes it from PIN syndrome

History

- Lateral proximal forearm pain
- Mimics lateral epicondylitis / "resistant tennis elbow"
- Classic: failed lateral epicondylitis treatment

Physical Exam

- Tenderness 3–4 cm **distal** to lateral epicondyle (not at the epicondyle)
- Pain with resisted middle finger extension (stresses ECRB over PIN)
- Pain with resisted forearm supination
- **No motor deficit**

Electrodiagnostics: Typically normal — diagnosis is clinical

Non-Surgical Management

- Activity modification, forearm splinting in neutral rotation
- NSAIDs
- Counterforce bracing (similar to tennis elbow strap, but positioned more distally)
- Prolonged conservative trial recommended — surgical outcomes are variable

Surgical: Decompress PIN through interval between BR and ECRL (Mackinnon); can also approach medial to BR or trans-BR. Release all 5 compression sites.

B. Posterior Interosseous Nerve (PIN) Syndrome

- PIN = deep branch of radial nerve, **pure motor**
- Innervates: supinator, EDC, ECU, EDM, APL, EPL, EPB, EIP
- ECRL/ECRB spared (branch proximal to PIN takeoff)

Presentation

- Finger drop (loss of MCP extension)
- Wrist extension present but with **radial deviation** (ECU out, ECRL intact)
- No sensory loss

Causes: Arcade of Frohse compression, lipoma, ganglion, RA synovitis, fracture

Non-Surgical: Observe if Parsonage-Turner suspected (up to 3 months); splint for function

Surgical: Decompression; get MRI first to rule out mass

C. Wartenberg Syndrome (Superficial Radial Nerve)

- Pure sensory — emerges between brachioradialis and ECRL ~9 cm proximal to radial styloid
- Burning pain/numbness over dorsoradial hand
- Aggravated by wrist motion, tight watchbands
- Can mimic de Quervain's (Finkelstein may be positive) — always check both
- **Management:** Remove offending compression, splinting, desensitization; surgical neurolysis rarely needed

V. Anomalous Neural Connections

Martin-Gruber Connection (forearm)

- Median-to-ulnar motor crossover in the proximal forearm (~15–20% of population)
- Median nerve fibers cross to ulnar nerve and innervate ulnar-typical intrinsic hand muscles
- Clinical relevance:
  - CTS EDX: may show initially positive CMAP deflection stimulating median at elbow (volume-conducted from ulnar muscles)
  - Can cause falsely fast median motor conduction velocities across the forearm
  - In high ulnar nerve lesions, hand intrinsic function may be partially spared (confusing exam)

Riche-Cannieu Connection (hand)

- Communication between recurrent motor branch of median nerve and deep branch of ulnar nerve in the hand (~35% of population)
- Clinical relevance:
  - In severe CTS, thenar muscles may be partially spared despite complete median nerve compression
  - In ulnar neuropathy, median nerve may partially compensate for intrinsic weakness
  - Can lead to "all ulnar" or "all median" hands in extreme variants
- **Teaching pearl:** When exam findings don't match the expected nerve distribution, consider anomalous connections before assuming the localization is wrong

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VI. Basic EMG/NCS Interpretation

Nerve Conduction Studies

Parameter	Measures	Abnormal →
Distal latency	Time to first response	Prolonged → distal demyelination
Conduction velocity	Speed across segment	Decreased → segmental demyelination
Amplitude (SNAP/CMAP)	Number of viable axons	Decreased → axonal loss
Conduction block	Amplitude drop across segment	Focal demyelination at that site

Needle EMG

Finding	Significance
Fibrillation potentials / PSWs	Active denervation
Decreased recruitment	Fewer motor units available
Polyphasic MUPs	Reinnervation (collateral sprouting) — chronic
Large-amplitude MUPs	Chronic reinnervation

Key Teaching Points

- **Sensory changes appear first** in entrapment — small myelinated fibers are most vulnerable
- **Motor changes** = more advanced compression
- **Fibrillations take ~3 weeks** to develop after axonal injury — timing of study matters
- **Severity framework:**
  - Mild: prolonged latency, normal amplitude (demyelination only → good prognosis)
  - Moderate: prolonged latency + mildly reduced amplitude

- Severe: markedly reduced amplitude, fibrillations on needle (axonal loss → guarded prognosis)

VII. Discussion Questions

1. Patient has median nerve distribution numbness, negative EMG, CTS-6 score of 5. Next steps?
2. How do you distinguish radial tunnel syndrome from lateral epicondylitis on exam?
3. Patient had CTR 6 months ago and symptoms returned. Walk through your workup.
4. You see fibrillation potentials in APB on needle EMG. What does this tell you about severity and timing?
5. Patient has finger drop, intact wrist extension with radial deviation, no sensory loss. Where is the lesion?